

A Survey of Graph Neural Networks and Graph Foundation Models for Scientific Discovery

InteractiveSurvey

Abstract

The accelerating pace of scientific discovery increasingly relies on extracting meaningful patterns from complex, non-Euclidean data, yet traditional machine learning often struggles with the intricate topological dependencies inherent in these domains. Graph Neural Networks have emerged as a transformative solution, though standard models frequently lack the scalability and broad semantic understanding required for universal scientific inquiry. This survey critically examines the convergence of Graph Neural Networks and Graph Foundation Models, exploring the paradigm shift from specialized architectures to large-scale, pre-trained systems capable of zero-shot adaptation. We synthesize recent advancements in unified architectural paradigms, self-supervised pre-training objectives, and novel adaptation frameworks designed to overcome data scarcity and domain shifts. Our analysis highlights how hybrid message-passing mechanisms, transformer-based tokenization strategies, and geometric encodings synergize to balance local structural fidelity with global contextual awareness. Furthermore, we evaluate contrastive, generative, and predictive pre-training methods that enable models to internalize fundamental physical laws rather than merely memorizing surface patterns. Finally, we discuss prompt-based unification techniques that facilitate flexible few-shot adaptation across diverse scientific tasks. By providing a holistic roadmap connecting architectural design, training methodologies, and deployment strategies, this work serves as a foundational reference for leveraging next-generation graph AI to drive breakthroughs in scientific understanding and innovation.

1 Introduction

The accelerating pace of scientific discovery increasingly relies on the ability to extract meaningful patterns from complex, non-Euclidean data

structures inherent in domains such as materials science, drug discovery, and particle physics. Traditional machine learning approaches often struggle to capture the intricate topological dependencies and long-range interactions present in these systems, limiting their predictive power and generalizability. Graph Neural Networks (GNNs) have emerged as a transformative solution, offering a principled framework for modeling relational data by leveraging message-passing mechanisms to aggregate local neighborhood information [1]. However, while standard GNNs excel at task-specific learning, they frequently lack the scalability and broad semantic understanding required to serve as universal engines for scientific inquiry across diverse and unseen domains.

This survey paper critically examines the convergence of Graph Neural Networks and Graph Foundation Models (GFMs) specifically tailored for scientific discovery. We explore the paradigm shift from specialized, small-scale GNNs to large-scale, pre-trained foundation models capable of zero-shot adaptation and cross-domain reasoning. The central research topic focuses on how architectural innovations, advanced pre-training strategies, and novel adaptation frameworks are synergizing to overcome the limitations of data scarcity and domain shift in scientific applications. By synthesizing recent advancements, this work aims to provide a comprehensive roadmap for developing robust graph-based AI systems that can accelerate hypothesis generation and experimental validation in the natural sciences.

The core of this survey begins with an in-depth analysis of unified architectural paradigms that define the backbone of modern graph foundation models. We investigate the evolution from strict message-passing schemes to hybrid architectures that integrate attention mechanisms, balancing local structural fidelity with global contextual awareness. Special attention is given to transformer-

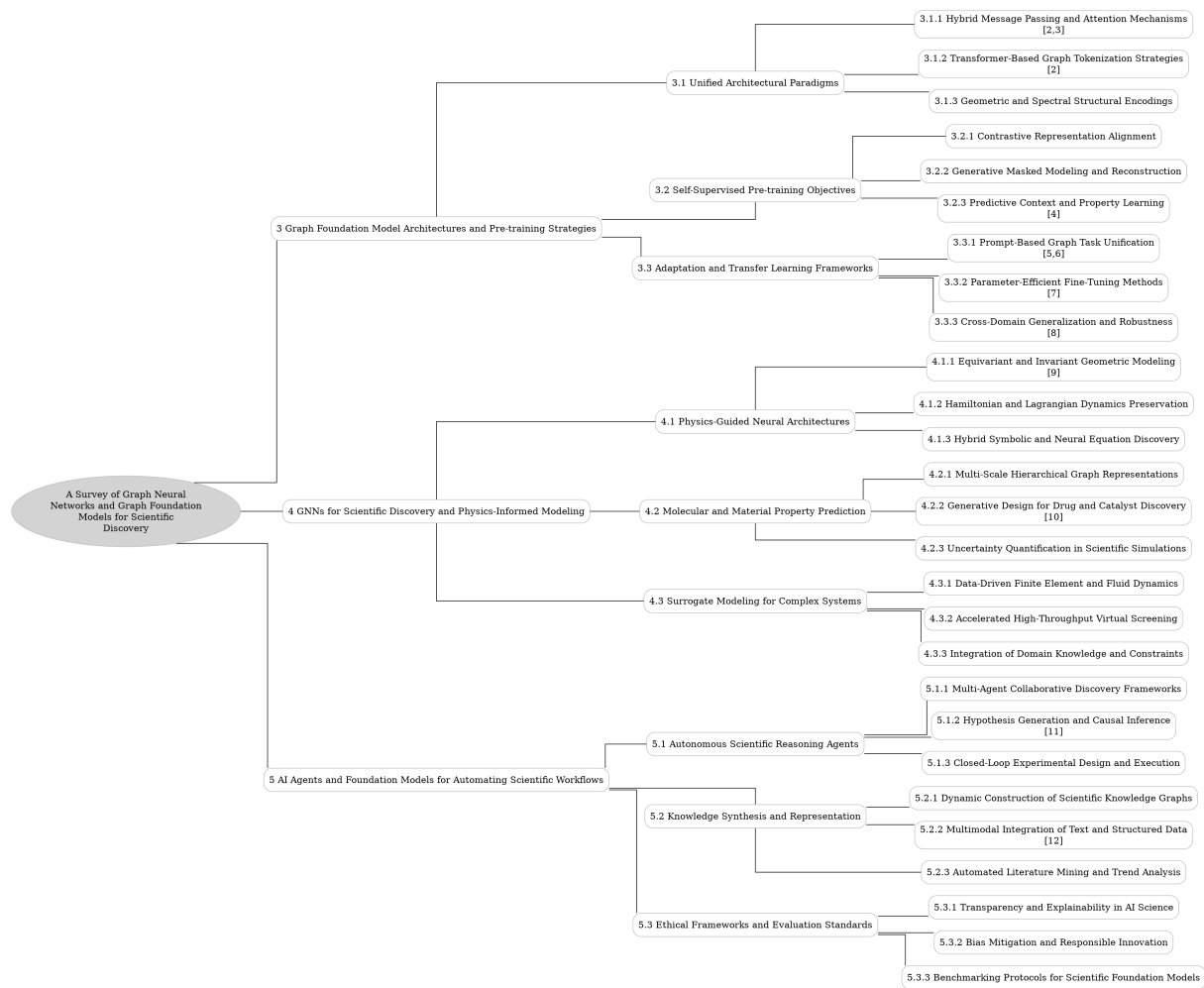


Figure 1: The outline of our survey: A Survey of Graph Neural Networks and Graph Foundation Models for Scientific Discovery

$$\min_{\theta, \omega} \frac{1}{|\mathcal{S}|} \sum_{v_i \in \mathcal{S}} \|g_{\omega}(\mathbf{h}_i) - \mathbf{x}_i\|^2$$

Figure 2: Chart from 'Self-supervised Learning on Graphs: Contrastive, Generative, or Predictive' (arXiv:2105.07342)

based tokenization strategies that convert irregular graph structures into ordered sequences, enabling the application of scalable self-attention [2]. Furthermore, we detail the critical role of geometric and spectral structural encodings, which inject non-Euclidean inductive biases into these models, ensuring they can effectively capture the complex manifold properties and long-range dependencies characteristic of scientific datasets.

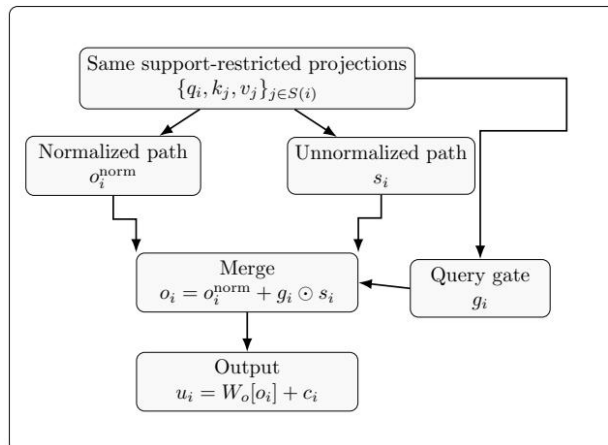


Figure 3: Chart from 'Cardinality-Preserving Attention Channels for Graph Transformers in Molecular Property Prediction' (arXiv:2602.02201)

Subsequently, the discussion transitions to self-supervised pre-training objectives that empower these models to learn transferable representations without extensive manual annotation. We categorize and evaluate three dominant paradigms: contrastive representation alignment, which maximizes mutual information across modalities; generative masked modeling, which forces the reconstruction of corrupted structural and feature components; and predictive context learning, which anticipates non-trivial properties to encode causal relationships. These strategies are essential for compressing heterogeneous scientific data into a unified latent space, allowing models to internalize fundamental physical and chemical laws rather than merely memorizing surface-level patterns.

Finally, the survey explores adaptation and transfer learning frameworks designed to bridge the gap between pre-training and diverse downstream scientific tasks. We examine prompt-based unification methods that reformulate distinct problems-ranging from node classification to link prediction-into a standardized text-to-text or token-to-token format. This section highlights how integrating learnable prompts with lightweight backbones facilitates flexible inference and few-shot adaptation, enabling a single foundation model to address multiple scientific challenges efficiently. By decoupling task-specific instructions from the core representation learning process, these frameworks significantly enhance the versatility and deployability of graph models in real-world research settings.

The primary contribution of this work lies in its systematic synthesis of the rapidly evolving landscape of graph foundation models for scientific applications. Unlike previous reviews that focus narrowly on algorithmic improvements or specific domains, this survey provides a holistic view connecting architectural design, pre-training methodologies, and adaptation techniques. We offer a critical assessment of current limitations and identify key opportunities for future research, particularly in enhancing interpretability and scaling to massive molecular and material graphs. Ultimately, this paper serves as a foundational reference for researchers aiming to leverage next-generation graph AI to drive breakthroughs in scientific understanding and innovation.

2 Graph Foundation Model Architectures and Pre-training Strategies

2.1 Unified Architectural Paradigms

2.1.1 Hybrid Message Passing and Attention Mechanisms

Hybrid message passing and attention mechanisms represent a pivotal evolution in graph representation learning, aiming to synthesize the structural inductive biases of Graph Neural Networks with the global receptive fields of Transformers [2]. Traditional Message Passing Neural Networks aggregate information strictly from local neighborhoods, which can limit long-range dependency modeling. Conversely, standard Graph Transformers apply self-attention globally but often struggle with scalability and may overlook

fine-grained topological nuances without explicit structural constraints. Hybrid architectures address these limitations by integrating attention modules directly into the message passing framework or restricting attention scopes to specific graph neighborhoods, thereby balancing local fidelity with global context awareness.

Recent methodological advancements have introduced sophisticated strategies to fuse these paradigms, such as employing attention-based aggregation functions within standard message passing steps or utilizing graph-structured masks to constrain self-attention operations. For instance, certain models revise the standard attention computation by incorporating spatial and edge encodings, ensuring that attention weights reflect both feature similarity and topological proximity. Other approaches deploy lightweight message-passing backbones alongside learnable prompts, where the backbone captures local relational patterns dependent on connectivity while attention mechanisms process explicit structural descriptions. This synergy allows the model to maintain meaningful representations across graphs with diverse feature spaces and varying topological complexities.

Furthermore, the integration of heterogeneous attention blocks and multi-head mechanisms has enhanced model expressivity without significantly increasing parameter counts [3]. By sharing attention matrices across different edge types or utilizing basis vectors for node-specific prompts, these hybrid models effectively capture complex interactions in non-regular distributions. The combination of flash attention for initial layers and memory-efficient implementations for deeper stacks further optimizes computational overhead, enabling the training of larger configurations on extensive graph datasets. Ultimately, these hybrid designs facilitate robust transfer learning and improve performance on tasks ranging from node classification to graph-level prediction by leveraging the complementary strengths of localized message propagation and global attentional weighting.

2.1.2 Transformer-Based Graph Tokenization Strategies

Transformer-based graph learning fundamentally relies on converting irregular graph structures into ordered token sequences compatible with standard attention mechanisms [2]. The most prevalent strategy treats individual nodes as discrete tokens, where raw node features are projected into a latent

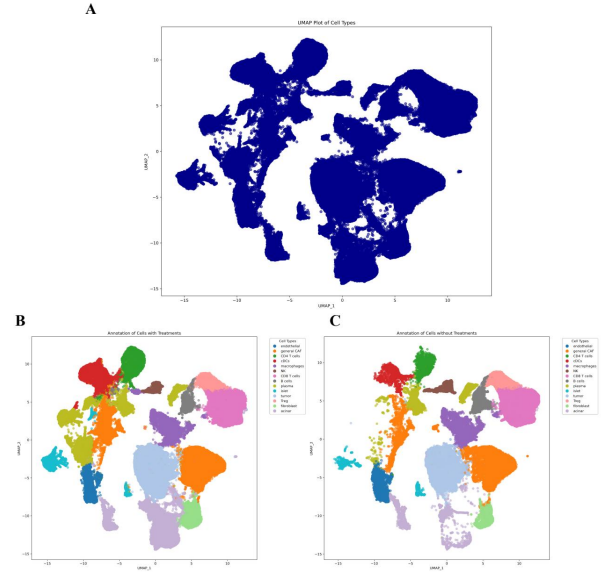


Figure 4: Chart from 'Large Language Models Meet Graph Neural Networks for TextNumeric Graph Reasoning' (arXiv:2501.16361)

space to form the initial input sequence. However, since vanilla transformers lack inherent inductive biases for topology, this naive tokenization must be augmented with explicit structural encodings. Techniques such as centrality, spatial, and edge encodings are injected into the attention bias or feature embeddings to preserve relational information, ensuring the model captures both local neighborhoods and global connectivity patterns without relying on message-passing schemes.

To address the scalability limitations imposed by the quadratic complexity of full-graph attention, advanced tokenization strategies have evolved beyond simple node-level serialization. Hierarchical approaches construct tokens from higher-order substructures, such as motifs, communities, or computation trees, effectively compressing the graph representation while retaining semantic richness. Methods leveraging level-order traversals or neighborhood aggregation generate composite tokens that encapsulate multi-hop contextual information, allowing the transformer to process large-scale graphs more efficiently. These strategies mitigate the memory bottlenecks associated with massive node counts by reducing the sequence length while maintaining the expressivity required for complex reasoning tasks.

Recent developments further integrate multi-modal information by aligning graph tokens with textual embeddings derived from large language models. In these frameworks, node descriptions

or structural profiles are encoded into semantic vectors that serve as soft tokens, bridging the gap between symbolic graph data and natural language paradigms. This alignment facilitates cross-domain pretraining and enables the unification of diverse graph tasks under a single text-to-text framework. By dynamically adapting token granularity-from fine-grained nodes to coarse-grained subgraphs-modern tokenization strategies provide the flexibility necessary for building generalizable graph foundation models capable of handling heterogeneous data distributions and varying graph sizes.

2.1.3 Geometric and Spectral Structural Encodings

Geometric and spectral structural encodings address the fundamental limitation of standard transformers in capturing non-Euclidean topological dependencies. Spectral methods leverage the graph Laplacian to inject global structural awareness, often utilizing eigenvectors as positional embeddings to capture low-frequency smooth patterns and long-range connectivity. To mitigate issues regarding sign and basis ambiguity inherent in spectral decomposition, recent advances employ sign-invariant networks or learnable scaling coefficients that allow the model to adaptively weigh spectral components against semantic node features, ensuring robust representation across varying graph scales.

Complementing spectral approaches, geometric encodings map graph structures into diverse manifold spaces, including hyperbolic and spherical geometries, to better preserve hierarchical and cyclic properties that Euclidean spaces fail to represent efficiently. By embedding nodes into product spaces combining multiple manifolds, models can adaptively learn the most suitable curvature for specific substructures, thereby enhancing the fidelity of local motif preservation. This geometric perspective is further reinforced by distance-biased random walks and spatial encoding mechanisms that explicitly inject centrality and shortest-path information into the attention bias, effectively grounding the latent space in the graph’s intrinsic metric properties.

To maximize expressive power, contemporary frameworks integrate these geometric and spectral priors with learnable structure-based mechanisms that break structural symmetries. Techniques such as random feature initialization com-

bined with equivariant processing ensure that the resulting embeddings remain invariant to node permutations while distinguishing between structurally similar but distinct topologies. By synthesizing fixed spectral bases, adaptive geometric manifolds, and stochastic structural markers, these encoding strategies provide a comprehensive multi-scale representation. This fusion enables foundation models to generalize across unseen graph domains, capturing both the fine-grained local geometry and the coarse-grained global spectral signature essential for complex downstream tasks.

2.2 Self-Supervised Pre-training Objectives

2.2.1 Contrastive Representation Alignment

Contrastive representation alignment serves as a pivotal mechanism for compressing heterogeneous graph structures into a unified latent space, effectively creating a shared vocabulary for cross-domain reasoning. By leveraging Perceiver Encoders with cross-attention mechanisms, raw domain-specific features and positional embeddings are projected into common latent tokens. This process facilitates the alignment of disparate modalities, such as graph topology and natural language descriptions, ensuring that semantically similar entities occupy proximal regions in the embedding hypersphere regardless of their original structural variations.

The core learning objective is formulated as a classification task within a minibatch, where the model identifies correct left-hand side and right-hand side pairings through contrastive loss functions. This approach maximizes the mutual information between positive pairs while pushing negative samples apart, thereby enforcing both alignment and uniformity on the hypersphere. Unlike reconstruction-based methods that may prioritize local fidelity, contrastive alignment explicitly optimizes for global semantic consistency, capturing both contextual relationships and intrinsic symbol information without relying on explicit data augmentation or negative mining in certain negative-free frameworks.

Advanced implementations extend this paradigm to handle multi-view inputs and spectral properties, integrating shift-robust invariant alignment losses to mitigate distribution shifts across datasets. By aligning graph encodings with summary representations or textual descriptors,

these methods enable transferable feature learning that transcends specific graph domains. The resulting representations preserve causal and structural integrity, allowing downstream tasks to leverage a robust, pre-aligned foundation that bridges the gap between discrete graph structures and continuous vector spaces essential for large-scale foundation models.

2.2.2 Generative Masked Modeling and Reconstruction

Generative masked modeling has emerged as a pivotal self-supervised paradigm for graph representation learning, adapting the success of masked language modeling to non-Euclidean domains. The core mechanism involves corrupting input graphs by masking a subset of node features, edge connections, or substructures, and subsequently training an encoder-decoder architecture to reconstruct these hidden components. Unlike contrastive methods that rely on instance discrimination, this generative approach forces the model to learn rich semantic and structural dependencies by minimizing reconstruction errors, such as mean squared error for continuous attributes or cross-entropy for categorical labels, thereby capturing intrinsic data distributions without manual annotation.

Recent advancements have diversified reconstruction targets beyond simple feature recovery to include complex structural patterns and multi-task objectives. Methods like GraphMAE and its variants focus on reconstructing masked node attributes using specialized decoders, while others such as EdgeMask and GPT-GNN target link prediction or autoregressive generation of graph elements. Sophisticated masking strategies, including random walks and anchor-based sub-path masking, are employed to create challenging pretext tasks that prevent trivial solutions. Furthermore, hybrid objectives often combine feature denoising with structure recovery, utilizing joint loss functions to ensure the learned embeddings encode both local attribute semantics and global topological information effectively.

The efficacy of generative masked modeling heavily relies on the design of the reconstruction head and the masking ratio, which balance the difficulty of the pretext task against the availability of contextual information. Decoupling aggregation and transformation layers, as seen in recent architectures, allows for more flexible message passing and improved gradient flow during the re-

construction phase. By optimizing these generative objectives, models achieve robust generalization across downstream tasks, demonstrating that high-fidelity reconstruction of masked graph components serves as a powerful proxy for learning transferable representations capable of handling domain shifts and sparse labeling scenarios.

2.2.3 Predictive Context and Property Learning

Predictive context and property learning constitutes a pivotal paradigm in graph foundation models, shifting focus from mere structural reconstruction to the anticipation of non-trivial node and graph attributes. Unlike traditional pretext tasks that rely on recovering masked features, this approach leverages self-supervised objectives to predict context-aware properties within local subgraphs or across k-hop neighborhoods. By formulating tasks such as link existence verification or attribute forecasting, models learn to encode intrinsic topological roles and semantic relationships, thereby capturing causal features rather than spurious correlations inherent in raw graph data.

A sophisticated extension of this paradigm involves hierarchical training curricula designed to refine supervision from coarse global signals to fine-grained local constraints [4]. Techniques such as random walk context reconstruction enable models to internalize path dependencies by masking interior nodes while observing endpoints, effectively teaching the network to infer missing structural information. This progressive strategy facilitates the learning of robust structural priors that are agnostic to specific node identities or feature schemas, allowing the model to generalize across diverse domains without requiring extensive task-specific fine-tuning.

$$\mathbf{Y}^* = \arg \min_{\mathbf{Y} \in \mathcal{F}(G, s)} J(\mathbf{Y}; G, s).$$

Figure 5: Chart from 'GRAPH FOUNDATION MODELS: BRIDGING LANGUAGE MODEL PARADIGMS AND GRAPH OPTIMIZATION' (arXiv:2509.24256)

Furthermore, predictive property learning serves as the mechanistic backbone for enabling in-context learning capabilities within graph neural networks. By training on auxiliary self-supervised objectives, such as predicting whether a node belongs to a specific contextual

neighborhood, models acquire the flexibility to adapt to novel tasks via prompting rather than parameter updates. This capability allows foundation models to map feature-agnostic graph properties into structured prompts, supporting zero-shot inference and few-shot adaptation where the model utilizes provided demonstrations to align its prediction space with new downstream objectives efficiently.

2.3 Adaptation and Transfer Learning Frameworks

2.3.1 Prompt-Based Graph Task Unification

Prompt-based graph task unification aims to transform heterogeneous graph learning objectives into a standardized prediction format, enabling a shared pretraining and inference pipeline [5]. This explicit unification strategy categorizes tasks by granularity into link-level, subgraph-level, and tree-level formulations, effectively bridging the gap between diverse downstream applications. By reformulating distinct problems such as node classification, link prediction, and graph clustering into a unified prompt-response paradigm, these methods facilitate the development of versatile graph foundation models capable of cross-task generalization without requiring task-specific architectural modifications.

$$\max_{\theta, \phi} \frac{1}{|\mathcal{D}|} \sum_{(\mathbf{A}, \mathbf{X}, y) \in \mathcal{D}} \sum_{i=1}^N \log P(y_i | g_{\phi}(f_{\theta}(\mathbf{A}, \mathbf{X})_i)),$$

Figure 6: Chart from 'One Prompt Fits All: Universal Graph Adaptation for Pretrained Models' (arXiv:2509.22416)

To achieve effective alignment, recent approaches leverage Large Language Models by converting graph structures into textual or structured serializations, such as XML-like markup languages or natural language descriptions centered on specific nodes. Advanced frameworks integrate learnable graph prompt tokens with lightweight message-passing backbones, allowing the model to capture both explicit structural descriptions and local relational patterns simultaneously [6]. This hybrid design ensures that the backbone learns representations that remain meaningful across graphs with varying feature spaces and topologies, while the prompts provide a flexible interface for injecting task-specific instructions and external knowledge sources.

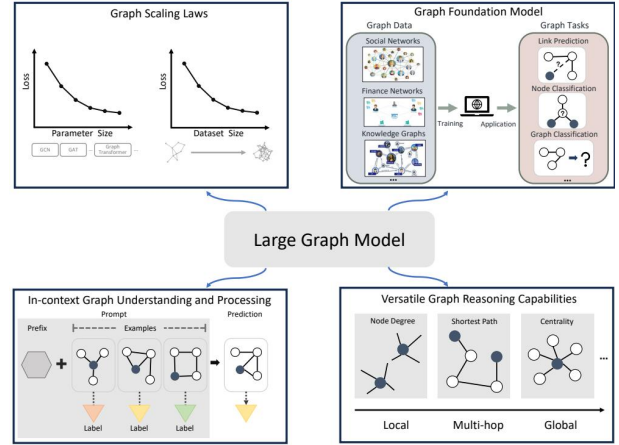


Figure 7: Chart from 'Graph Meets LLMs: Towards Large Graph Models' (arXiv:2308.14522)

Despite these advancements, significant challenges persist regarding the optimality and interpretability of manually designed prompts, which often struggle to preserve intrinsic graph inductive biases during textual conversion. Consequently, research has shifted towards automatically generated prompts utilizing LLM summarization capabilities and soft prompt tuning to mitigate human bias and enhance adaptability. While current unified templates predominantly address discriminative tasks, emerging efforts are expanding this scope to include generative and open-ended objectives, striving to establish a comprehensive "text-to-graph" framework that matches the versatility of natural language foundation models.

2.3.2 Parameter-Efficient Fine-Tuning Methods

Parameter-Efficient Fine-Tuning (PEFT) has emerged as a critical paradigm for adapting large-scale pre-trained graph models to downstream tasks while mitigating the computational burdens and catastrophic forgetting associated with full parameter updates. Unlike standard fine-tuning, which optimizes the entire parameter set θ , PEFT strategies freeze the backbone architecture and introduce lightweight, trainable modules. This approach preserves the generalizable representations learned during pre-training, ensuring robustness across diverse graph structures while significantly reducing memory footprint and training latency [7].

Prominent techniques in this domain include adapter-based methods and low-rank transformations tailored for graph neural networks. Approaches such as AdapterGNN and G-Adapter in-

$$\arg \min_C \sum_{k=1}^K \sum_{\mathbf{h}_i \in C_k} \|\mathbf{h}_i - \boldsymbol{\mu}_k\|_2^2.$$

Figure 8: Chart from 'GraphSculptor: Sculpting Pre-training Coreset for Graph Self-supervised Learning' (arXiv:2605.01310)

tegrate compact adapter modules, typically structured as down-projection, non-linear activation, and up-projection layers with batch normalization, directly into GNN layers. These modules enable task-specific adaptation by updating only a small fraction of parameters. Similarly, Low-Rank Adaptation (LoRA) constrains weight updates to low-rank decomposition matrices, offering a flexible mechanism to tune specific sub-modules or type-indexed matrices without altering the original pre-trained weights.

The optimization of PEFT methods often employs advanced schedulers and regularizers to stabilize convergence within limited epochs. Common configurations utilize AdamW optimizers with cosine decay schedulers and warmup periods to navigate the loss landscape effectively. By decoupling the adaptation process from the massive backbone, PEFT facilitates rapid deployment on consumer-grade hardware and enhances performance in data-scarce regimes. Consequently, these methods strike an optimal balance between maintaining the model’s foundational knowledge and achieving high accuracy on specialized graph tasks.

2.3.3 Cross-Domain Generalization and Robustness

Achieving robust cross-domain generalization in graph foundation models remains a formidable challenge due to the intrinsic heterogeneity of graph data across different fields. Unlike vision or language domains, graphs from distinct sources often lack unified underlying semantics, shared feature schemas, or common node identities, leading to significant distribution shifts that degrade performance. This structural and semantic disparity frequently results in weak transfer or even negative transfer when models pretrained on one domain, such as molecular structures, are applied to another, like knowledge graphs or road networks, without explicit alignment mechanisms.

To mitigate these domain gaps, recent ap-

proaches emphasize learning transferable structural priors rather than relying solely on domain-specific features. Techniques such as domain regularization, adversarial training, and topology alignment project disparate graphs into a shared embedding space, effectively minimizing the distribution gap between pretraining and fine-tuning data [8]. By focusing on invariant topological characteristics-such as random walk patterns or subgraph motifs-models can capture universal graph properties that persist across domains, thereby preventing catastrophic forgetting and enabling adaptability to unseen distributions while maintaining task-specific accuracy.

Empirical analyses reveal that generalization performance is highly sensitive to both model scale and the diversity of the pretraining corpus. While increasing model parameters and token counts generally improves out-of-distribution accuracy, the benefit is contingent upon the compatibility of feature encodings across source and target domains. Transfers between domains with aligned feature spaces show marked improvement, whereas those involving divergent schemas require sophisticated normalization or text-free alignment strategies. Consequently, future advancements must balance architectural expressivity with robust regularization to ensure stable performance under large shifts in graph structure, sparsity, and semantic context.

3 GNNs for Scientific Discovery and Physics-Informed Modeling

3.1 Physics-Guided Neural Architectures

3.1.1 Equivariant and Invariant Geometric Modeling

Equivariant and invariant geometric modeling establishes the mathematical foundation for processing data with inherent symmetries, particularly within Euclidean and Riemannian spaces. Invariant Graph Neural Networks (GNNs) utilize scalar features, such as inter-atomic distances and bond angles, to ensure that predicted properties remain unchanged under translation, rotation, and permutation operations [9]. Conversely, equivariant models extend this capability by processing tensorial features, guaranteeing that the output transforms consistently with the input coordinates. This distinction is critical in domains like materials chemistry and molecular dynamics, where preserving physical laws regarding force fields and energy

landscapes requires strict adherence to symmetry constraints.

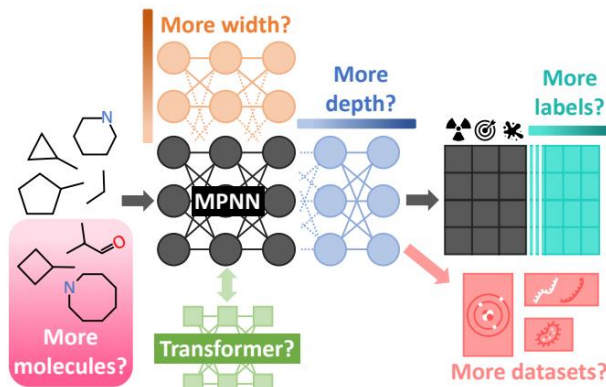


Figure 9: Chart from 'On the Scalability of GNNs for Molecular Graphs' (arXiv:2404.11568)

Recent architectural advancements have shifted from purely invariant formulations to sophisticated equivariant frameworks, such as $E(n)$ -GNNs and Geometric Algebra Transformers, which operate directly on coordinate vectors. These models leverage message-passing mechanisms that update both scalar and vector representations simultaneously, enabling the capture of higher-order geometric interactions without relying on hand-crafted invariant descriptors. By incorporating Riemannian geometry, these approaches address the limitations of Euclidean projections, allowing for coordinate-free computations of curvature and connection. This facilitates more intuitive analysis of intrinsic manifold structures and improves the modeling of complex distributions through techniques like Riemannian flow matching.

Theoretical analyses demonstrate that equivariant models generally exhibit superior accuracy and transferability compared to their invariant counterparts, especially when dealing with sparse data or distributional shifts. Double equivariance frameworks further enhance robustness by enforcing symmetry constraints on both node features and relation types, ensuring stable convergence in tasks ranging from linker design to force field prediction. While invariant methods effectively mitigate the curse of dimensionality by discarding non-essential geometric noise, they risk feature collapse; equivariant architectures resolve this by maintaining rich, distance-preserving latent representations that align with the underlying physical dynamics of the system.

3.1.2 Hamiltonian and Lagrangian Dynamics Preservation

Preserving the symplectic structure of physical systems is paramount for long-term stability in learned dynamical models. Hamiltonian Graph Neural Networks address this by parameterizing the total energy of an N -body system as a scalar function of node positions and momenta, rather than directly learning acceleration vectors. By deriving equations of motion through automatic differentiation of this learned Hamiltonian, these architectures inherently satisfy Liouville's theorem, ensuring that phase space volume is conserved during integration. This approach effectively mitigates the energy drift commonly observed in standard black-box neural ODE solvers, allowing for accurate trajectory prediction over extended time horizons without explicit regularization constraints.

Complementary to the Hamiltonian formalism, Lagrangian dynamics preservation leverages the principle of least action to model system evolution through the difference between kinetic and potential energy. In this framework, graph-based representations encode generalized coordinates, where the Lagrangian is constructed to be invariant under specific symmetry groups such as rotation and translation. The resulting Euler-Lagrange equations are then solved using specialized numerical integrators that respect the underlying variational structure. This method proves particularly advantageous for systems with complex holonomic constraints, as the geometric properties of the configuration manifold are naturally embedded within the message-passing mechanism, facilitating robust generalization across varying topological configurations.

Recent advancements have extended these physics-informed architectures to incorporate dissipative forces and external potentials while maintaining numerical stability. By coupling symplectic integrators with learnable Rayleigh dissipation functions, modern frameworks can simulate non-conservative processes without violating the core geometric invariants of the conservative backbone. Furthermore, the integration of meta-learning strategies enables rapid adaptation of these dynamic priors to unseen physical domains, enhancing cross-domain generalization. Despite these successes, computational overhead remains a challenge, as the requirement for higher-

order derivatives and implicit integration schemes increases inference latency compared to standard explicit solvers, necessitating ongoing research into efficient approximation techniques.

3.1.3 Hybrid Symbolic and Neural Equation Discovery

Hybrid symbolic and neural equation discovery represents a paradigm shift in scientific machine learning, aiming to reconcile the data-driven flexibility of deep learning with the interpretability and physical consistency of symbolic regression. Traditional neural networks often function as black boxes, struggling to generalize outside their training distributions or adhere to known physical laws. By integrating symbolic reasoning modules directly into neural architectures, these hybrid models can explicitly search for closed-form mathematical expressions that govern underlying system dynamics. This approach leverages the differentiable nature of neural components to optimize symbolic trees, enabling the discovery of governing equations that are both accurate and human-readable.

The architectural implementation of these systems typically involves a neuro-symbolic loop where a neural network proposes candidate equation structures while a symbolic engine refines their parameters and functional forms. In this framework, graph neural networks frequently serve as the backbone for encoding relational data, such as molecular graphs or particle interactions, into latent representations. These embeddings are then decoded into symbolic operators and constants through attention mechanisms or reinforcement learning policies. The loss function is often augmented with constraints derived from dimensional analysis or conservation laws, ensuring that the discovered equations respect fundamental physical principles without requiring exhaustive labeled data for every possible state.

Recent advancements have demonstrated the efficacy of this hybridization in domains ranging from quantum chemistry to fluid dynamics, where precise mechanistic understanding is paramount. Unlike pure neural surrogates that approximate input-output mappings, hybrid models can extrapolate to unseen regimes by identifying invariant relationships within the data. Furthermore, the integration of pre-trained language models for symbolic token generation has accelerated the search space exploration, allowing for the rapid identifica-

tion of complex non-linear differential equations. As these methods mature, they promise to automate the formulation of scientific theories, bridging the gap between raw observational data and formalizable scientific knowledge.

3.2 Molecular and Material Property Prediction

3.2.1 Multi-Scale Hierarchical Graph Representations

Multi-scale hierarchical graph representations address the limitation of standard message-passing mechanisms in capturing long-range dependencies and global structural patterns. By mimicking the pooling operations found in convolutional neural networks, these architectures progressively coarsen the graph topology to generate embeddings at varying levels of granularity. Differentiable pooling techniques enable the model to learn optimal cluster assignments dynamically, effectively identifying discriminative subgraphs while preserving essential topological information across scales.

The construction of these hierarchies often relies on adaptive structure-aware mechanisms that aggregate node features into super-nodes, thereby reducing computational complexity for large-scale networks. This process facilitates the extraction of both local neighborhood characteristics and broader community structures within a unified framework. Through iterative layers of aggregation and downsampling, the model builds a pyramid of representations where higher layers encode increasingly abstract semantic concepts, analogous to visual feature hierarchies in image processing.

Furthermore, integrating hierarchical pooling with attention-based transformers enhances the expressiveness of graph foundation models by combining local connectivity with global context. Such hybrid architectures mitigate the over-smoothing problem common in deep graph neural networks by maintaining distinct feature distributions at each scale. The resulting multi-resolution embeddings prove particularly effective for graph-level tasks, such as molecular property prediction and social network analysis, where understanding interactions across different structural scopes is critical for accurate inference.

3.2.2 Generative Design for Drug and Catalyst Discovery

Generative design has emerged as a pivotal paradigm for accelerating the discovery of novel therapeutics and catalytic materials by navigating vast chemical spaces that are intractable for traditional high-throughput screening. By leveraging deep generative models, researchers can perform inverse design, specifying desired physicochemical properties to synthesize candidate molecules *de novo*. This approach significantly reduces the temporal and financial costs associated with experimental synthesis, enabling the rapid iteration of molecular scaffolds with optimized binding affinities, stability, and selectivity for specific biological targets or catalytic reactions.

A critical advancement in this domain involves the integration of Graph Neural Networks (GNNs) within generative frameworks to explicitly model the topological and geometric constraints of molecular structures [10]. Unlike sequence-based representations, GNNs capture the intricate connectivity and spatial relationships inherent in bio-chemical graphs, facilitating the generation of chemically valid and synthetically accessible compounds [10]. Recent methodologies combine these graph-based architectures with reinforcement learning and variational autoencoders to optimize multi-objective functions, ensuring that generated candidates not only exhibit high predicted activity but also adhere to strict drug-likeness criteria and reaction feasibility rules.

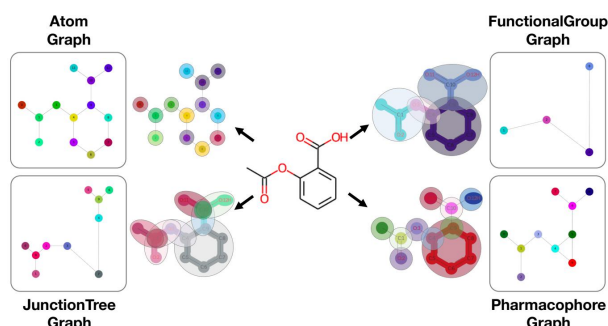


Figure 10: Chart from 'Enhancing Model Learning and Interpretation using Multiple Molecular Graph Representations for Compound Property and Activity Prediction' (arXiv:2304.06253)

Despite these successes, significant challenges remain regarding data distributional shifts and the reliable prediction of properties for novel scaffolds outside the training manifold. Current efforts are

increasingly focusing on physics-informed neural networks that embed quantum-chemical principles to enhance the fidelity of interaction modeling, particularly for complex catalyst-adsorbate systems. Furthermore, the development of robust benchmarks capable of evaluating model performance under realistic distributional shifts is essential to ensure the generalizability of generative designs, paving the way for their deployment in automated, self-driving laboratory environments for end-to-end scientific discovery.

3.2.3 Uncertainty Quantification in Scientific Simulations

Uncertainty quantification (UQ) in scientific simulations addresses the critical challenge of modeling epistemic and aleatoric uncertainties where high-fidelity experimental data is scarce or noisy. In domains such as drug-target interaction prediction and materials discovery, Bayesian neural networks and heteroscedastic Gaussian noise models are increasingly employed to capture residual uncertainty and prevent overconfident failures. These probabilistic frameworks allow models to distinguish between inherent stochasticity in physical processes, such as Brownian motion, and uncertainties arising from limited training distributions, thereby enhancing the reliability of predictions in low-data regimes.

To handle the complex geometric constraints inherent in molecular and physical systems, advanced UQ methods integrate geometry-aware inference with manifold learning techniques. Approaches utilizing Symmetric Positive Definite submanifolds and Geodesic Random Walks enable approximate sampling via stochastic differential equations, where noise is generated in tangent spaces and mapped back to the manifold through exponential maps. This variational inference strategy ensures that uncertainty estimates respect the underlying topological structure of the data, facilitating stable flows and accurate side-chain packing in biomolecular simulations while avoiding the pitfalls of Euclidean approximations in non-linear configuration spaces.

Despite algorithmic advancements, significant gaps remain in calibrating uncertainty for out-of-distribution extrapolation and interpreting failure modes in large-scale simulations. Current metrics, including Mean Squared Error and modified R^2 , often fail to fully penalize deviations in high-dimensional property spaces, necessitating the de-

velopment of distance-aware deterministic methods and ensemble-based calibration techniques. Future research must prioritize the tight integration of these geometric frameworks with mechanistic interpretability, ensuring that uncertainty estimates not only quantify prediction confidence but also guide efficient exploration of vast, unconstrained experimental spaces for robust scientific discovery.

3.3 Surrogate Modeling for Complex Systems

3.3.1 Data-Driven Finite Element and Fluid Dynamics

Data-driven approaches are increasingly supplanting traditional discretization schemes in finite element analysis and computational fluid dynamics by learning solution operators directly from high-fidelity simulations. Unlike conventional methods that rely on rigid mesh generation and iterative solvers, graph neural networks and operator learning frameworks map spatial domains to continuous solution fields, effectively approximating partial differential equations without explicit numerical integration. These models leverage large-scale datasets generated from density functional theory and direct numerical simulations to capture complex non-linearities and boundary conditions that often challenge mechanistic solvers.

A critical advancement in this domain involves the integration of geometric deep learning with physics-informed constraints to ensure physical consistency. By encoding conservation laws, such as mass and momentum preservation, directly into the loss function or network architecture, these hybrid models mitigate the risk of unphysical predictions common in purely data-driven regimes. Specifically, equivariant architectures have demonstrated superior performance in modeling fluid-structure interactions and turbulent flows, as they inherently respect the rotational and translational symmetries of the underlying physical systems while reducing the sample complexity required for training.

Despite these successes, significant challenges remain regarding scalability and generalization to unseen geometries or extreme parameter regimes. Current methodologies often struggle with the computational overhead of training on high-resolution three-dimensional meshes and the sensitivity to noise in experimental sensor data. Future

research directions emphasize the development of foundation models capable of multi-fidelity learning, where coarse simulations are corrected by sparse high-fidelity data, and the adoption of adaptive mesh refinement strategies guided by learned error estimators to optimize computational resources dynamically.

3.3.2 Accelerated High-Throughput Virtual Screening

Accelerated high-throughput virtual screening (HTVS) increasingly relies on Graph Neural Networks (GNNs) implemented within optimized deep learning frameworks to process vast chemical libraries efficiently. By leveraging GPU-accelerated backends such as PyTorch Geometric, modern architectures utilize specialized mechanisms like Flash Attention to minimize the computational footprint during message passing. This hardware-aware optimization enables the rapid extraction of molecular fingerprints and embeddings, transforming the screening of millions of compounds from a prohibitive computational burden into a tractable task suitable for interactive diagnostic workflows and large-scale deployment.

To further enhance throughput, scalable training strategies employing distributed data parallelism across multiple nodes have become essential for handling expansive datasets. Techniques such as caching frozen base model embeddings on disk significantly reduce memory consumption and compute costs during downstream probing, allowing for efficient fine-tuning without recalculating gradients for the entire network. Furthermore, the integration of synthetic pretraining tasks and surrogate modeling facilitates the approximation of complex physical properties, enabling models to generalize effectively across diverse topologies while maintaining linear scaling in training throughput as computational resources expand.

Despite these advancements, balancing speed with rigorous validation remains a critical challenge in HTVS pipelines. While two-stage frameworks effectively reduce the search space through rapid candidate filtering, the subsequent verification stage often necessitates resource-intensive code execution or physics-based simulations that can bottleneck the overall workflow. Future optimizations must therefore focus on developing unified evaluation protocols and robust surrogate models that accurately predict binding affinities and ADMET properties without relying solely

on expensive first-principles calculations, ensuring that acceleration does not compromise the reliability of identified drug candidates.

3.3.3 Integration of Domain Knowledge and Constraints

The integration of domain knowledge into foundation models addresses the inherent limitations of purely data-driven approaches, particularly regarding task ambiguity and the scarcity of labeled scientific data. By embedding structured component effects, chemically motivated interaction terms, and expert-informed failure rules directly into the model architecture, systems can capture core mechanistic behaviors that generic learners often miss. This strategy shifts the modeling paradigm from simulating raw reaction dynamics to predicting endpoint behaviors governed by known physical or chemical constraints, thereby reducing the search space for valid solutions and enhancing interpretability.

Technically, this integration is frequently realized through the modification of message-passing algorithms within graph neural networks to respect predefined graph structures and relational priors. Methods such as multi-curvature spaces allow for the precise embedding of attribute values and temporal evolution, ensuring that structural information is not omitted during representation learning. Furthermore, incorporating hard constraints via custom loss functions or architectural biases prevents modality collapse and ensures that generated artifacts adhere to fundamental scientific laws, such as conservation principles or specific motif preservation in molecular graphs.

Despite these advancements, challenges remain in balancing dimension reduction with information preservation to avoid feature collapse under distributional shifts. Effective frameworks must navigate the trade-off between discarding noisy data and retaining critical domain-specific features necessary for robust generalization. Future directions point toward simulation-pretrained models that leverage principled null models for evaluation, enabling low-supervision learning that maintains high fidelity to domain expertise while scaling across diverse scientific tasks without requiring exhaustive retraining for every new application.

4 AI Agents and Foundation Models for Automating Scientific Workflows

4.1 Autonomous Scientific Reasoning Agents

4.1.1 Multi-Agent Collaborative Discovery Frameworks

Multi-agent collaborative discovery frameworks represent a paradigm shift from static, template-driven pipelines to dynamic, society-level simulations of scientific inquiry. These systems model research communities as heterogeneous graphs where autonomous agents assume distinct roles, such as literature review, hypothesis generation, experimental design, and peer review. By orchestrating interactions through graph-based communication protocols, these frameworks facilitate iterative knowledge refinement and enable the emergence of novel insights that often elude individual agents or rigid workflows. The core architecture typically leverages large language models as reasoning engines capable of planning complex sequences, invoking external tools for simulation or data retrieval, and synthesizing multimodal inputs ranging from textual descriptions to atomic structures.

A critical advancement in this domain is the integration of structured knowledge bases and ontologies to ground agent reasoning in domain-specific facts, thereby mitigating hallucinations and enhancing reproducibility. Frameworks increasingly employ agentic tree search and cognitive taxonomy guidance to explore vast solution spaces, systematically targeting underexplored behavioral patterns rather than relying solely on pre-trained distributions. This approach allows for the automated discovery of causal relationships and the optimization of synthesis procedures in fields like materials science and computational fluid dynamics. Furthermore, the inclusion of human-in-the-loop mechanisms ensures that expert intuition guides the collaborative process, balancing autonomous exploration with validated scientific constraints.

Despite significant progress, challenges remain in scaling these systems to handle the full complexity of real-world scientific discovery, particularly regarding the interoperability of diverse data modalities and the evaluation of long-term collaborative dynamics. Current benchmarks often lack the breadth to assess causal graph discovery and probabilistic reasoning across interdisciplinary contexts. Future developments aim to es-

establish standardized evaluation metrics and decentralized ecosystems that prevent over-optimization while fostering robust, self-correcting research societies. Ultimately, these frameworks promise to accelerate the scientific cycle by automating routine cognitive tasks and enabling researchers to focus on high-level strategic direction and creative synthesis.

4.1.2 Hypothesis Generation and Causal Inference

Hypothesis generation in modern scientific discovery increasingly leverages large language models to navigate vast combinatorial spaces of concepts, shifting the researcher's role from ideation to judgment [11]. As automation lowers the marginal cost of generating candidate hypotheses, the relative economic and cognitive returns to human judgment intensify, necessitating systems that can effectively filter trivial or redundant proposals. Agentic frameworks now facilitate this by constructing initial ideation traces through semantic random walks, producing candidate questions with supporting evidence that are subsequently stress-tested in zero-shot settings to ensure robustness before human evaluation.

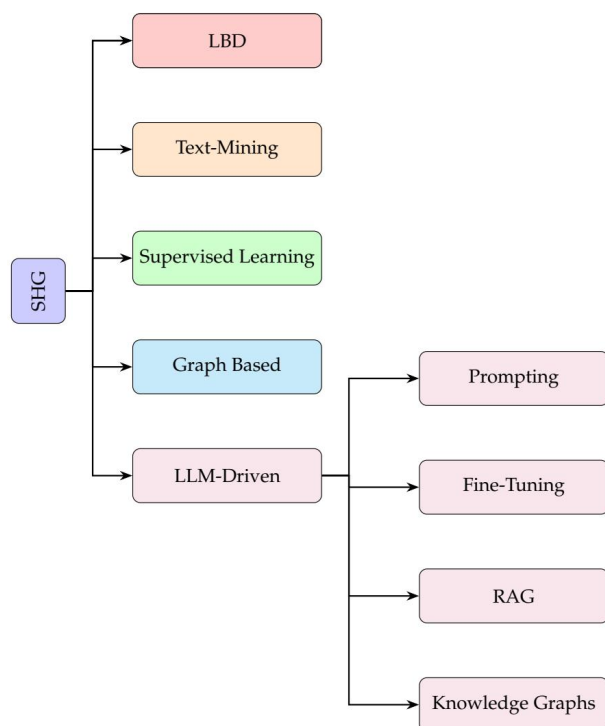


Figure 11: Chart from 'A Survey on Hypothesis Generation for Scientific Discovery in the Era of Large Language Models' (arXiv:2504.05496)

To transition from mere correlation to action-

able scientific insight, these generated hypotheses must be grounded in rigorous causal inference mechanisms rather than statistical association alone. Contemporary approaches construct methodological graphs from large-scale corpora, where citation networks are reified into typed causal edges representing lineage, extension, and generalization relations. By mapping paper-level interactions to method-level dependencies, researchers can perform lineage reconstruction to trace causal ancestors and evaluate novel ideas based on their structural position within the existing knowledge graph, thereby disentangling spurious correlations from genuine causal mechanisms.

Despite these advances, significant challenges remain in ensuring the veracity and reproducibility of machine-generated causal claims, particularly when foundation models introduce unquantified uncertainties into the inductive inference pipeline. The integration of Bayesian estimation techniques offers a pragmatic pathway to confirm or disconfirm hypotheses based on available evidence without asserting absolute truth, yet standard confidence intervals often fail to capture uncertainties arising from human judgment calls embedded in model training. Consequently, future frameworks must prioritize explainability and provenance tracking, treating causal discovery not as a black-box prediction task but as a transparent, iterative process of hypothesis refinement supported by verifiable data structures.

4.1.3 Closed-Loop Experimental Design and Execution

Closed-loop experimental design represents a paradigm shift from static hypothesis testing to dynamic, iterative discovery driven by real-time feedback. In this framework, artificial intelligence agents do not merely execute pre-defined protocols but actively analyze incoming data streams to refine subsequent experimental parameters. This approach integrates uncertainty-aware exploration with multi-property optimization, allowing the system to navigate complex, rugged payoff spaces more efficiently than traditional high-throughput screening methods. By continuously updating probabilistic models of the search space, the system can identify promising regions for synthesis or simulation that human researchers might overlook due to cognitive biases or dimensional constraints.

The execution of these closed loops relies on robust automation architectures capable of seam-

less interaction between computational planning and physical or simulated actuation. Workflow management systems orchestrate the cycle of suggesting precursors, tuning process conditions, and guiding robotic experimentation without manual intervention. Critical to this process is the establishment of rigorous feedback mechanisms where experimental outcomes directly inform the refinement of computational surrogates. This ensures that the models adapt to real-world constraints and noise, thereby enhancing the dependability and liveness of the autonomous discovery process while minimizing the latency between technique development and scientific application.

Despite these advancements, the authenticity and rationality of closed-loop systems remain contingent upon careful environment construction and human supervision. Researchers must define specific application scenarios and boundary conditions to prevent the system from optimizing for spurious correlations or non-verifiable tasks. The integration of numerical simulations often exhibits a two-phase adoption curve, requiring a maturation period before methods become widely applicable across domains. Consequently, successful deployment demands a balance between full autonomy and expert oversight, ensuring that the generated designs are not only novel but also scientifically valid and reproducible within the broader context of the domain's theoretical framework.

4.2 Knowledge Synthesis and Representation

4.2.1 Dynamic Construction of Scientific Knowledge Graphs

The dynamic construction of scientific knowledge graphs represents a paradigm shift from static citation networks to evolving, heterogeneous structures that capture the temporal unfolding of research. Unlike traditional models that rely on fixed metadata, contemporary approaches leverage large language models to continuously extract entities, synthesis protocols, and causal relationships from raw textual corpora. This process transforms unstructured literature into queryable, machine-readable formats, enabling the real-time integration of new findings regarding material properties, experimental designs, and methodological innovations as they emerge in the scientific record.

Central to this methodology is the encoding of reasoning traces as heterogeneous transition

graphs, where nodes represent distinct conceptual elements and edges denote hierarchical, sequential, or parallel dependencies. By computing edge weights through normalized pointwise mutual information, these systems quantify the strength of associations between research tasks, effectively mapping the prototypical reasoning structures employed by successful scientific inquiries. This granular representation allows for the distinction between mere co-occurrence and functional dependency, facilitating a deeper understanding of how specific experimental actions lead to validated discoveries across diverse domains.

Furthermore, the dynamic nature of these graphs supports automated idea generation and the simulation of methodological evolution. As AI agents reallocate researcher effort from initial hypothesis formation to the evaluation of model-suggested candidates, the knowledge graph serves as a foundational data layer for iterative refinement. By modeling the research community as a living network, these systems can uncover latent interdisciplinary connections and predict emerging trends, thereby accelerating the discovery process through continuous structural updates that reflect the current state of scientific knowledge rather than historical snapshots.

4.2.2 Multimodal Integration of Text and Structured Data

The integration of unstructured textual narratives with rigid structured data represents a critical frontier in scientific AI, necessitating architectures that transcend simple concatenation. Modern approaches employ unified tokenization schemes and modality adapters to bridge disparate formats, transforming raw PDFs, tables, and molecular graphs into coherent semantic flow graphs. By leveraging layout parsing and metadata extraction, these systems preserve the intricate relationships between equations, reaction mechanisms, and tabular values, ensuring that the latent knowledge embedded in complex scientific literature is not lost during digitization.

To achieve robust fusion, contemporary models utilize joint representation learning modules that simultaneously encode node text and structural properties derived from citation graphs or chemical topologies. This dual-encoding strategy allows transformation functions to discover latent topics correlated with hidden factors across both modalities, effectively mapping papers and entities into a

shared vector space. Unlike traditional binary classifiers, these multimodal foundation models, such as Uni-SMART, demonstrate superior capability in interpreting heterogeneous inputs ranging from bar plots to detailed synthesis protocols, thereby overcoming the limitations of text-dominant reasoning.

Despite these advancements, challenges remain regarding the diversity of reasoning exhibited by multimodal models compared to their text-only counterparts [12]. Current evaluations suggest that while integration improves accuracy in specific downstream tasks like material composition design, there is a risk of reduced reasoning diversity when models rely heavily on visual cues without sufficient semantic grounding. Future research must therefore focus on developing sophisticated generative frameworks that balance text and image contributions, ensuring that the fusion components genuinely enhance understanding rather than merely aligning superficial features across modalities.

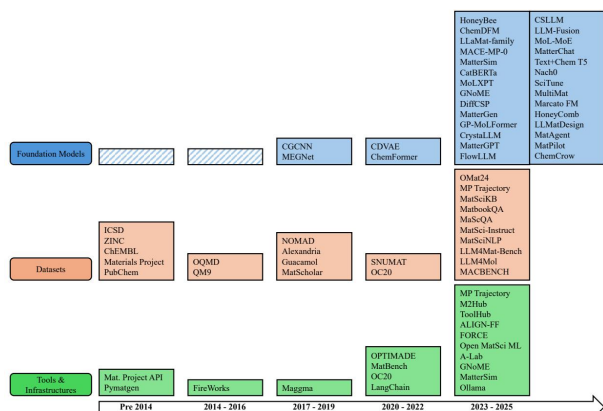


Figure 12: Chart from 'A SURVEY OF AI FOR MATERIALS SCIENCE: FOUNDATION MODELS, LLM AGENTS, DATASETS, AND TOOLS' (arXiv:2506.20743)

4.2.3 Automated Literature Mining and Trend Analysis

Automated literature mining has evolved from simple metadata aggregation to sophisticated, multimodal extraction systems capable of parsing full-text scientific articles. Traditional approaches relying on citation networks and keyword co-occurrence are increasingly supplemented by large language models fine-tuned on domain-specific corpora, enabling precise named entity recognition and the extraction of complex synthesis protocols. These systems address the challenge of un-

structured data by transforming heterogeneous inputs, including text, tables, and figures, into structured, queryable knowledge graphs that capture methodological evolution rather than mere document relationships.

The integration of these mining capabilities facilitates dynamic trend analysis and the automation of early-stage discovery tasks. By leveraging semantic similarity and topic modeling, AI agents can now systematically identify emerging technological trajectories and cluster publications based on latent conceptual relationships rather than explicit citations. This shift allows for the reallocation of human researcher effort from broad idea generation to the critical evaluation of model-suggested candidates, effectively creating a prioritized search framework where algorithms handle the initial screening of vast literature landscapes to highlight high-potential research directions.

Despite these advancements, significant barriers remain in achieving fully autonomous meta-scientific reasoning, primarily due to the scarcity of open-access full-text data and the difficulty of extracting precise logical inferences from natural language. Current methodologies often struggle with the nuance of hidden citations and the verification of extracted claims without extensive human curation. Future developments must focus on enhancing the interoperability of research data standards and improving the robustness of automated validation mechanisms to ensure that mined insights are both computationally useful and scientifically rigorous, thereby supporting the next generation of AI-driven research agents.

4.3 Ethical Frameworks and Evaluation Standards

4.3.1 Transparency and Explainability in AI Science

The integration of foundation models into scientific workflows has precipitated a critical demand for transparency, as the opaque nature of black-box architectures threatens the credibility of empirical findings. While these systems excel at internalizing vast domain knowledge to accelerate discovery, their reliance on hidden citations and non-interpretable latent representations creates significant information asymmetries. Consequently, the scientific community faces a reproducibility crisis where the provenance of generated hypotheses remains obscured, necessitating a paradigm shift

from purely predictive accuracy toward rigorous auditability and open infrastructure that tracks the full causal lineage of knowledge.

To mitigate these risks, contemporary research advocates for the development of Explainable AI (XAI) frameworks specifically tailored for scientific reasoning, moving beyond post-hoc explanations to inherently interpretable model designs. This involves enhancing uncertainty quantification to handle adversarial conditions and label noise, thereby providing trustworthy confidence intervals for predictions in high-stakes domains like healthcare and materials science. Furthermore, the deployment of semantic flow graphs and deterministic evaluation metrics allows researchers to trace the logical abstraction levels within reasoning traces, ensuring that models consistently generalize from concrete instances to abstract principles without hallucinating unsupported connections.

Achieving true transparency also requires a cultural and technical commitment to open science practices, including the release of model weights, training data provenance, and detailed documentation of prompt engineering strategies. Future directions emphasize the creation of pluralistic benchmark landscapes that evaluate not only task performance but also the robustness and interpretability of scientific agents under varying conditions. By embedding human oversight mechanisms and adhering to formalized standards for computability and stability, the field can foster the trust necessary for deploying autonomous systems in hypothesis-driven design and interactive scientific exploration.

4.3.2 Bias Mitigation and Responsible Innovation

Bias mitigation in scientific AI necessitates a multi-layered approach addressing both data provenance and algorithmic architecture. Since training datasets often encode historical disparities regarding gender, geography, and economic status, relying solely on raw data leads to unjust conclusions and skewed impact metrics. Effective strategies involve embedding detection mechanisms directly into the pipeline, utilizing techniques such as synthetic minority over-sampling to correct class imbalances and employing multi-model ensembles to dilute individual model biases. Furthermore, moving beyond traditional citation counts toward diversified evaluation metrics is essential to prevent the amplification of existing in-

equities in credit allocation.

Responsible innovation extends beyond technical corrections to encompass rigorous governance frameworks and transparency standards. Implementing veridical data science principles requires documenting all judgment calls, from data curation and preprocessing to hyperparameter tuning, ensuring that downstream users can audit decision-making pathways. The adoption of model cards and standardized reporting protocols facilitates accountability by explicitly detailing system limitations, potential risks, and the reasoning behind automated decisions. This transparency is critical for regulators and researchers to delineate responsibility, particularly when AI systems influence high-stakes scientific workflows or policy compliance.

Ultimately, sustaining healthy innovation ecosystems demands a shift from purely autonomous deployment to human-in-the-loop systems that prioritize interpretability and safety. While AI enhances the value of expert judgment, it must augment rather than replace human oversight to mitigate risks associated with opaque reasoning processes. Broad participation in developing assessment standards through interdisciplinary committees ensures that evaluative measures capture diverse perspectives on robustness and fairness. By integrating ethical constraints into the core design of foundation models, the scientific community can foster trustworthy tools that support reproducible research while minimizing the potential for misleading claims or systemic discrimination.

4.3.3 Benchmarking Protocols for Scientific Foundation Models

Establishing rigorous benchmarking protocols for scientific foundation models requires moving beyond static text completion tasks toward dynamic, multi-modal evaluation frameworks that mirror the iterative nature of discovery. Effective protocols must assess a model's capacity to navigate the full scientific workflow, from hypothesis generation and literature synthesis to experimental design and data interpretation. This necessitates benchmarks that integrate heterogeneous data sources, including structured knowledge graphs, unstructured textual corpora, and raw experimental outputs, to verify whether models can truly synthesize cross-domain insights rather than merely retrieving isolated facts.

A critical component of these protocols is the

validation of generated hypotheses against ground-truth physical laws and experimental constraints. Advanced evaluation suites increasingly incorporate automated feedback loops where model predictions are subjected to computational screening, such as density functional theory calculations or molecular dynamics simulations, to determine material stability and synthesis feasibility. By embedding these verification steps directly into the benchmarking pipeline, researchers can quantify the "ideation-execution gap," distinguishing between theoretically novel but practically impossible suggestions and viable scientific contributions that withstand rigorous computational or empirical scrutiny.

Furthermore, robust protocols must address the challenges of reproducibility and provenance inherent in autonomous scientific agents. Evaluations should track the lineage of generated insights, ensuring that models correctly attribute sources and maintain logical consistency across complex reasoning chains. Future benchmarking standards will likely evolve to include collaborative, extensible frameworks that support distributed validation across diverse scientific domains, fostering a culture of transparency. Such infrastructure is essential for certifying that foundation models possess the generalization capabilities required for real-world deployment, ultimately shifting the focus from superficial performance metrics to reliable, actionable scientific advancement.

5 Future Directions

Despite significant advancements in Graph Foundation Models (GFM) for scientific discovery, several critical limitations persist that hinder their widespread adoption as universal engines for inquiry. Current architectures often struggle to balance the computational efficiency required for massive-scale molecular and material graphs with the expressivity needed to capture complex, long-range physical interactions. While hybrid message-passing and attention mechanisms offer a promising middle ground, they frequently incur prohibitive memory costs when scaling to systems with millions of nodes, limiting their applicability to realistic scientific simulations. Furthermore, existing self-supervised pre-training objectives, whether contrastive or generative, tend to optimize for surface-level structural fidelity rather than deep causal understanding of underly-

ing physical laws. This results in models that may perform well on benchmark tasks but fail to generalize robustly to out-of-distribution scenarios or unseen domains where data scarcity is acute. Additionally, the interpretability of these black-box systems remains insufficient for high-stakes scientific applications, where understanding the mechanistic rationale behind a prediction is as crucial as the prediction itself.

Future research must prioritize the development of scalable, geometry-aware architectures that explicitly encode physical symmetries and conservation laws directly into the model backbone. Moving beyond standard Euclidean assumptions, next-generation GFMs should leverage advanced geometric deep learning techniques to operate natively on diverse manifolds, ensuring that predictions respect fundamental principles such as rotation equivariance and energy conservation. Concurrently, there is an urgent need to innovate pre-training strategies that move from purely statistical reconstruction to physics-informed generative modeling. By integrating differential equations and domain-specific constraints into the loss functions, models can be guided to learn representations that reflect true causal mechanisms rather than spurious correlations. Another vital direction involves the creation of unified, multimodal adaptation frameworks that seamlessly integrate graph structures with textual scientific knowledge and experimental metadata. Developing sophisticated prompt-tuning and instruction-following capabilities will enable a single foundation model to flexibly address diverse tasks, from property prediction to de novo molecule generation, with minimal task-specific fine-tuning.

The realization of these research directions holds the potential to fundamentally transform the landscape of scientific discovery. By overcoming current scalability and generalizability barriers, robust Graph Foundation Models can serve as autonomous collaborators in the laboratory, accelerating the hypothesis-generation cycle and reducing the reliance on costly, time-consuming experiments. Enhanced interpretability will foster greater trust among domain experts, facilitating the deployment of AI-driven insights in critical areas such as drug design, renewable energy material development, and climate modeling. Ultimately, bridging the gap between data-driven pattern recognition and first-principles scientific reasoning will unlock new frontiers in understand-

ing complex natural phenomena, leading to faster breakthroughs and more sustainable technological innovations.

6 Conclusion

This survey has systematically synthesized the rapid convergence of Graph Neural Networks and Graph Foundation Models, highlighting a decisive paradigm shift from specialized, small-scale architectures to universal, pre-trained engines capable of cross-domain scientific reasoning. We examined how unified architectural paradigms, particularly hybrid message-passing and attention mechanisms, effectively balance local structural fidelity with global contextual awareness, overcoming the inherent limitations of traditional models in capturing long-range dependencies. Furthermore, our analysis detailed the critical role of advanced tokenization strategies and geometric-spectral encodings in transforming irregular, non-Euclidean data into formats amenable to scalable self-attention. By evaluating dominant self-supervised pre-training objectives, including contrastive alignment and generative masked modeling, we demonstrated how these methods compress heterogeneous scientific data into unified latent spaces that internalize fundamental physical and chemical laws. Finally, the discussion on adaptation frameworks illustrated how prompt-based unification and lightweight fine-tuning enable zero-shot and few-shot learning, bridging the gap between massive pre-training and diverse downstream tasks in materials science, drug discovery, and particle physics.

The significance of this work lies in its holistic integration of architectural design, pre-training methodologies, and adaptation techniques into a single, coherent roadmap for the future of scientific AI. Unlike previous reviews that often isolate algorithmic improvements or focus narrowly on specific domains, this survey provides a comprehensive landscape that connects theoretical innovations with practical applicability in real-world research settings. By critically assessing the synergy between transformer-based scalability and graph-specific inductive biases, we offer a foundational reference for developing robust systems that transcend data scarcity and domain shift. This synthesis not only clarifies the current state of the art but also identifies the essential components required to build next-generation models capable of

accelerating hypothesis generation and experimental validation. Ultimately, this work underscores the potential of graph foundation models to serve as universal solvers that can generalize across unseen scientific challenges, thereby transforming the pace and scope of discovery in the natural sciences.

Looking forward, the trajectory of graph foundation models demands a concerted research effort to address remaining challenges in interpretability, scalability, and causal reasoning. While current architectures show promise, future developments must prioritize mechanisms that provide transparent insights into model predictions, ensuring trustworthiness in high-stakes scientific applications. Researchers are called to innovate beyond mere performance metrics, focusing instead on creating models that can explicitly reason about causal relationships and adhere to known physical constraints. There is an urgent need to scale these systems to handle massive molecular and material graphs without compromising computational efficiency, potentially through novel sparse attention mechanisms or hierarchical processing units. We encourage the community to foster interdisciplinary collaboration, combining expertise in machine learning, domain-specific sciences, and high-performance computing to drive the next wave of breakthroughs. By embracing these directions, the field can realize the full potential of graph-based AI as a transformative force in unlocking the secrets of the universe and solving humanity’s most pressing scientific problems.

References

- [1] Self-supervised Learning on Graphs Contrastive, Generative, or Predictive
- [2] Cardinality-Preserving Attention Channels for Graph Transformers in Molecular Property Prediction
- [3] Large Language Models Meet Graph Neural Networks for Text-Numeric Graph Reasoning
- [4] GRAPH FOUNDATION MODELS BRIDGING LANGUAGE MODEL PARADIGMS AND GRAPH OPTIMIZATION
- [5] One Prompt Fits All Universal Graph Adaptation for Pretrained Models
- [6] Graph Meets LLMs Towards Large Graph Models
- [7] GraphSculptor Sculpting Pre-training Coreset for Graph Self-supervised Learning

- [8] Out-of-Distribution Generalization in Graph Foundation Models
- [9] On the Scalability of GNNs for Molecular Graphs
- [10] Enhancing Model Learning and Interpretation using Multiple Molecular Graph Representations for Compound Property and Activity Prediction
- [11] A Survey on Hypothesis Generation for Scientific Discovery in the Era of Large Language Models
- [12] A SURVEY OF AI FOR MATERIALS SCIENCE FOUNDATION MODELS, LLM AGENTS, DATASETS, AND TOOLS